pfi

Python for Fixed Income

COURSE 2 · BOND MATH

Clean and Dirty Price

WEEK 2 · DAY 3

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The price between coupons

Buy 1,000,000 of face of Quorvane at 5.40 percent, settling 30 November 2026

What price is quoted, and how much cash leaves the account?

Two things change from week 1

1. The next coupon is four months away: part of a period
2. Two months of interest belong to the seller: accrued

Today builds on day 7's accrued interest: clean price plus accrued is what the buyer pays.

Today

8.1 The quote and the invoice

8.2 Part of a period, w = DSC / E

8.3 price and dirty_price

8.4 On a coupon date: week 1

8.5 The final coupon period

8.6 yield_to_maturity

8.7 The September blotter

The quote and the invoice

99.361551

clean price per 100: the quote, Excel's PRICE

0.875000

accrued interest per 100: day 7

100.236551

dirty price per 100: what the buyer pays

On 1,000,000 of face	Dollars
Principal: face x clean / 100	993,615.51
Accrued: face x accrued / 100	8,750.00
Invoice: principal + accrued	1,002,365.51

Excel side by side (tested in Excel)

`=PRICE(s,m,5.25/100,5.4/100,100,2,0)`	99.361551
`=PRICE(...)+100*5.25/100/2*COUPDAYBS(...)/COUPDAYS(...)`	100.236551

s = DATE(2026,11,30), m = DATE(2031,9,30); the same s, m, 2, 0 in COUPDAYBS and COUPDAYS

The ticket shows clean. Settlement moves the dirty amount.

Part of a period: w = DSC / E

```
w = DSC / E = 120 / 180 = 0.6667
```

k	Coupon date	Periods	Cash flow	PV
0	2027-03-31	0.667	2.625	2.578788
1	2027-09-30	1.667	2.625	2.510991
…	…	…	…	…
9	2031-09-30	9.667	102.625	79.324172

Sum of the PVs: 100.236551, the dirty price. Less accrued: 99.361551, PRICE.

Discount first, subtract accrued second.

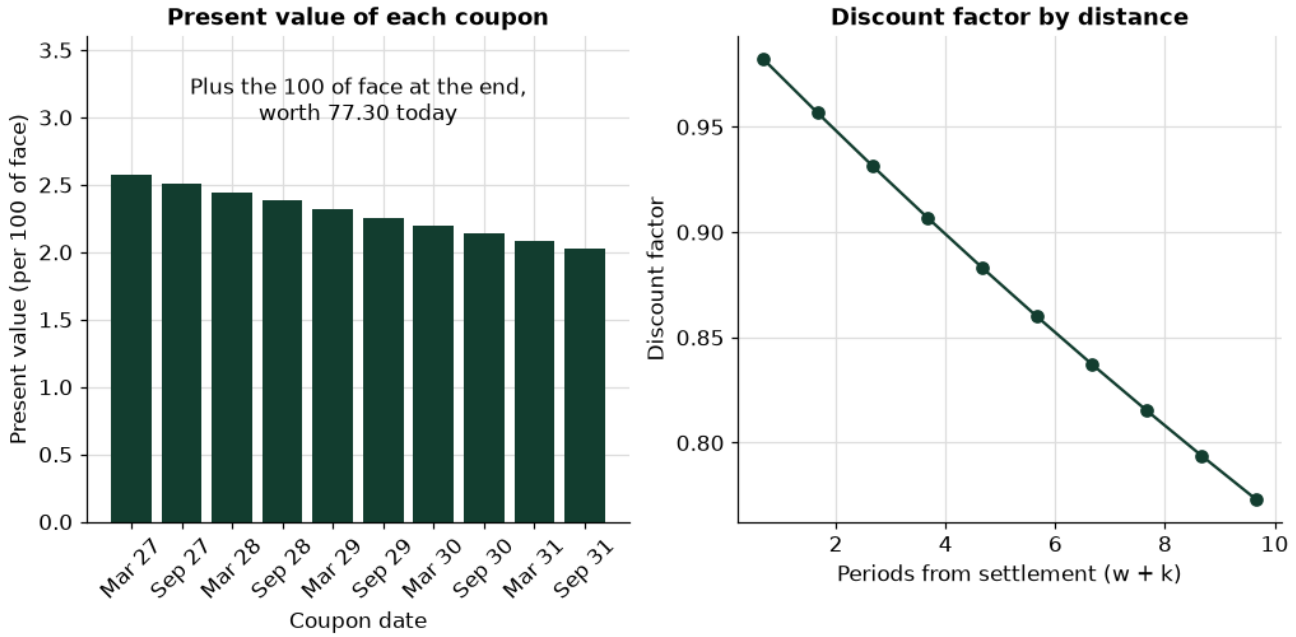
Excel side by side (tested in Excel)

`B1: =COUPDAYBS(s,m,2,0)`	60
`B2: =COUPDAYS(s,m,2,0)`	180
`B3: =COUPNUM(s,m,2,0)`	10
`B4: =(B2-B1)/B2`	w
`B6 down: =IF(A6=$B$3-1,...)/(1+y/2)^($B$4+A6)`	PV
`B16: =SUM(B6:B15)`	100.236551

One cell: =-PV(y/2,COUPNUM(...),c/2,100)*(1+y/2)^(COUPDAYBS/COUPDAYS) gives the same dirty price.

s = DATE(2026,11,30), m = DATE(2031,9,30), y = 5.4/100, c = 5.25

Discounting from settlement, drawn



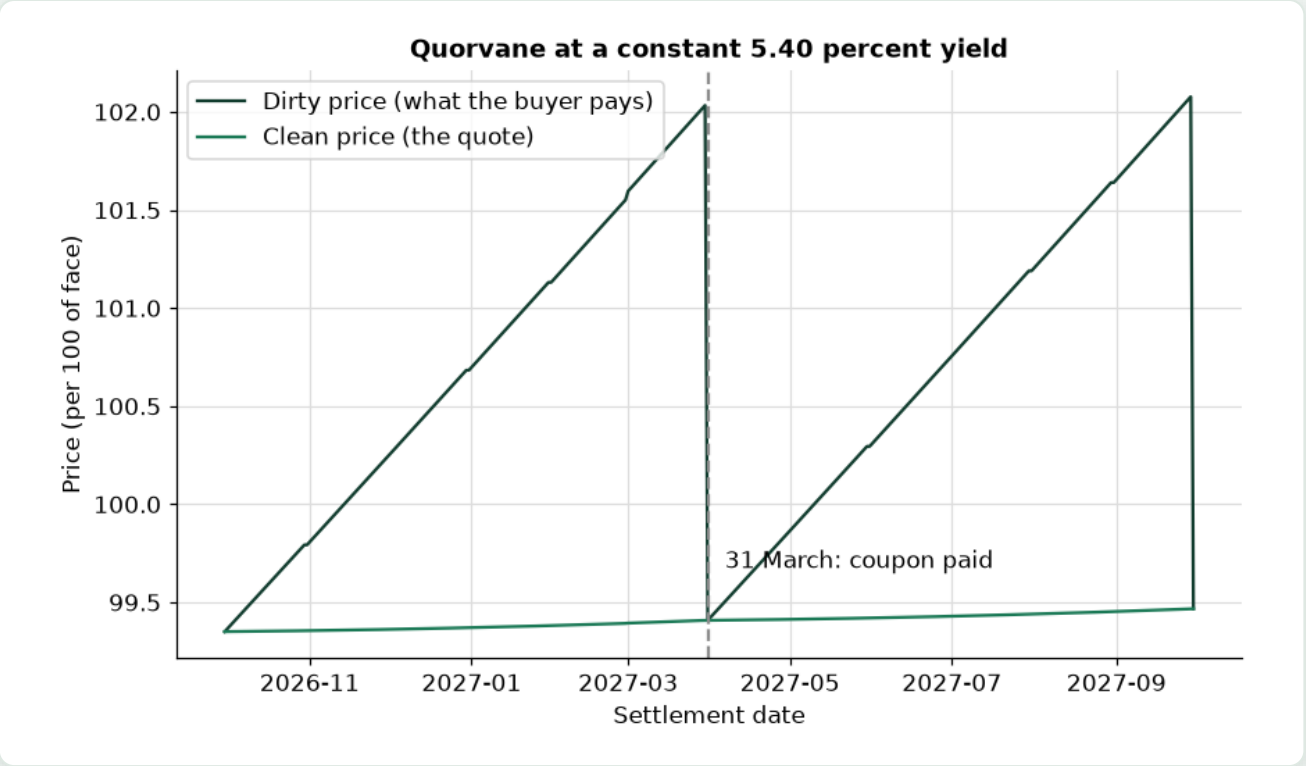
Each coupon is worth a little less the further away it is. The face at the end is worth the most: 77.30 today.

The discount factor falls steadily from the first coupon, w of a period away, to the last.

price and dirty_price, and why quotes are clean

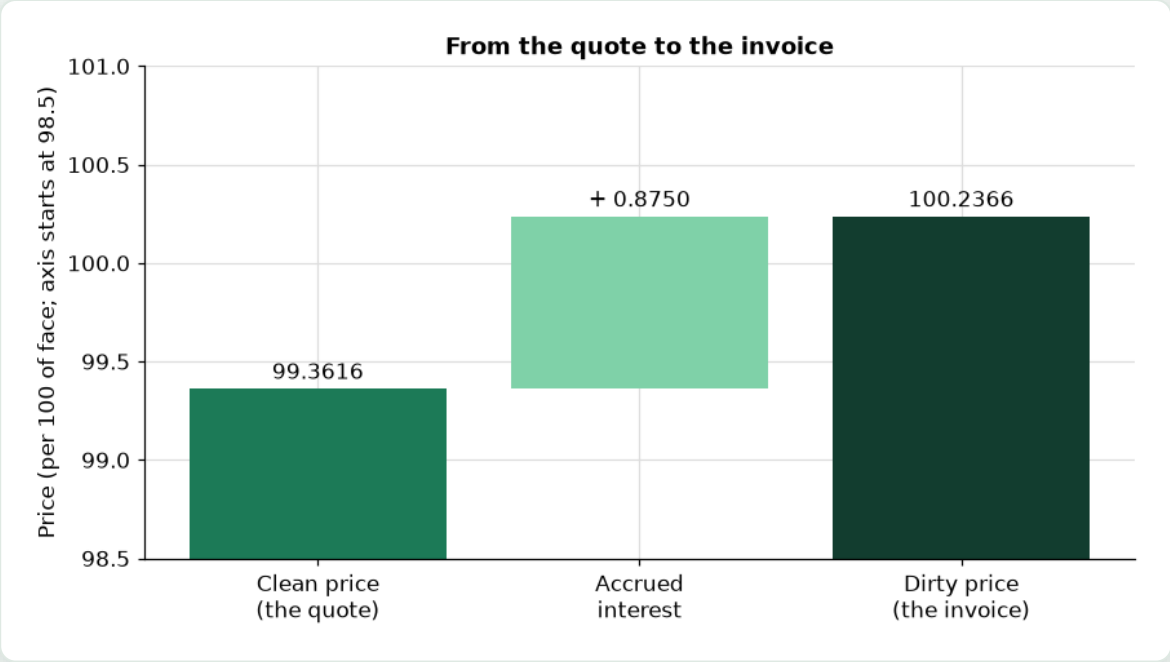
```
bondmath.price(settlement, maturity,
    coupon_pct, yield_pct)
# clean price per 100, as PRICE
bondmath.dirty_price(...)
# price + accrued_interest
```

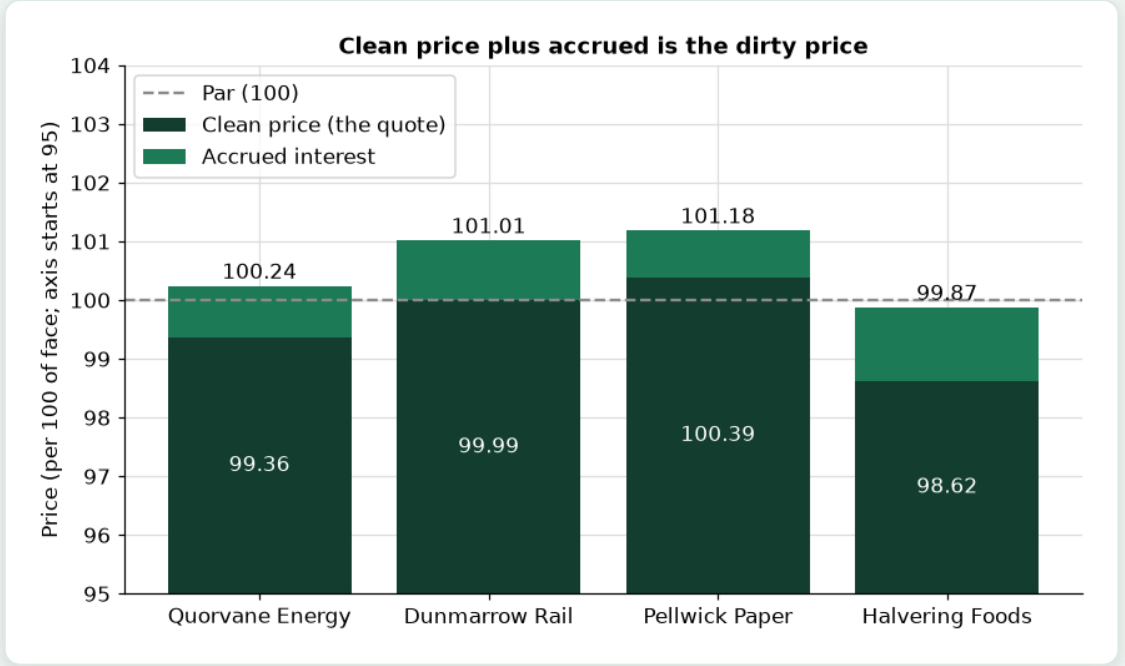
Bond	Clean	Dirty
Quorvane Energy	99.3616	100.2366
Dunmarrow Rail	99.9898	101.0106
Pellwick Paper	100.3882	101.1799
Halvering Foods	98.6225	99.8725



Dirty climbs, then drops a coupon. Clean moves smoothly: a change in the quote is a change in value.

From the quote to the invoice, drawn





The quote plus the interest the seller earned is the invoice price.

Dark is the quote, light is accrued, the top is the dirty price. The dashed line is par.

On a coupon date it equals week 1

A = 0

nothing accrued on 30 September 2026

w = 1

the next coupon one whole period away

The new function contains the old one as a special case. The old answers must not move.

Bond	price, 30 Sep	Excel -PV, week 1
Quorvane Energy	99.350327	99.350327
Dunmarrow Rail	100.000000	100.000000
Pellwick Paper	100.415887	100.415887
Halvering Foods	98.622736	98.622736

price equals price_from_yield to the twelfth decimal; the test checks every reference row that settles on a coupon date.

The final coupon period: simple interest

One coupon left: Excel's PRICE

```
dirty = (100 + c/f) / (1 + w x y/f)
```

Every other period compounds

```
(1 + y/f) ** (w + k)
```

Row edge_final_period: 5 percent coupon, 6 percent yield, settled 30 September 2026, maturing 15 February 2027, w = 0.75.

99.619499

simple interest: Excel's PRICE and bondmath.price

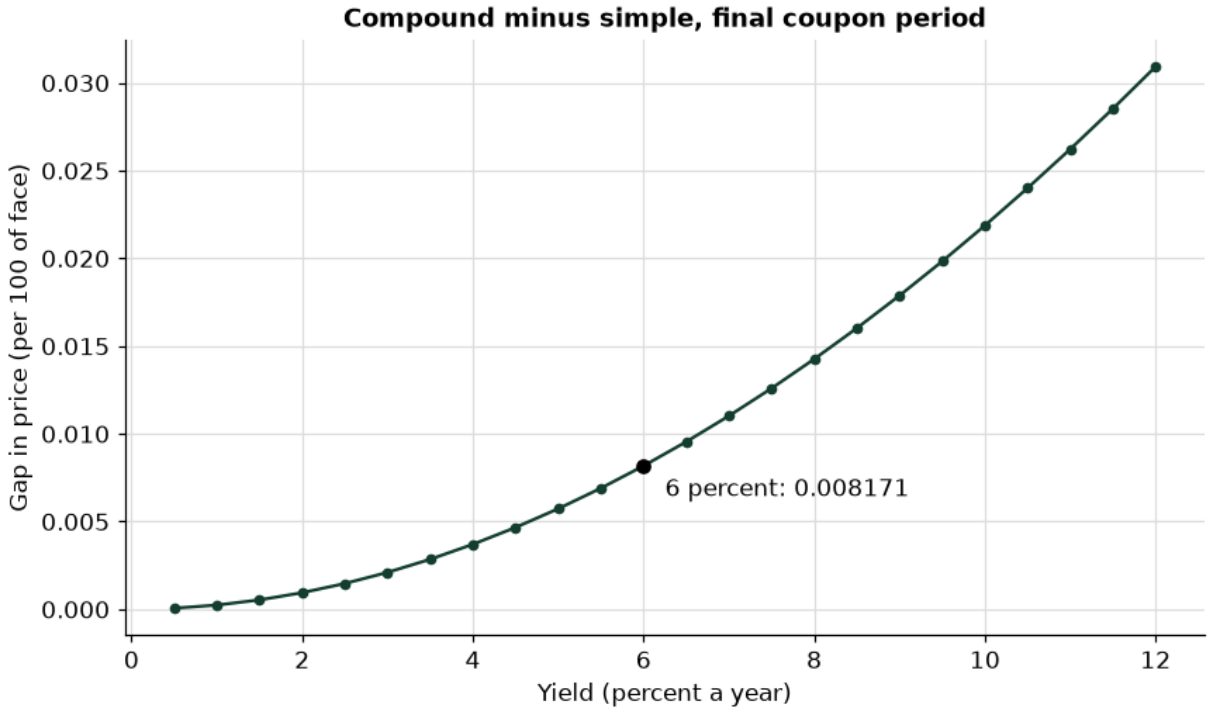
99.627670

compound, for comparison

Gap: 0.008171 per 100, 81.71 dollars on 1,000,000 of face. Tested on all final-period rows of the reference file.

Neither is wrong. The convention decides, and bondmath follows Excel.

Simple against compound, drawn



The conventions always differ, and the gap grows with the yield. Excel's PRICE uses simple interest, and so does bondmath.

yield_to_maturity, as Excel's YIELD

Bond	Quote	Yield 30 Sep	Yield 30 Nov
Quorvane Energy	99.125	5.4523	5.4565
Dunmarrow Rail	100.500	6.0364	6.0328
Pellwick Paper	100.875	4.4353	4.4159
Halvering Foods	97.750	7.8286	7.8300

Same clean price, two months on: every yield moves away from its coupon. A description of the formula, not a forecast.

Final period near a month end

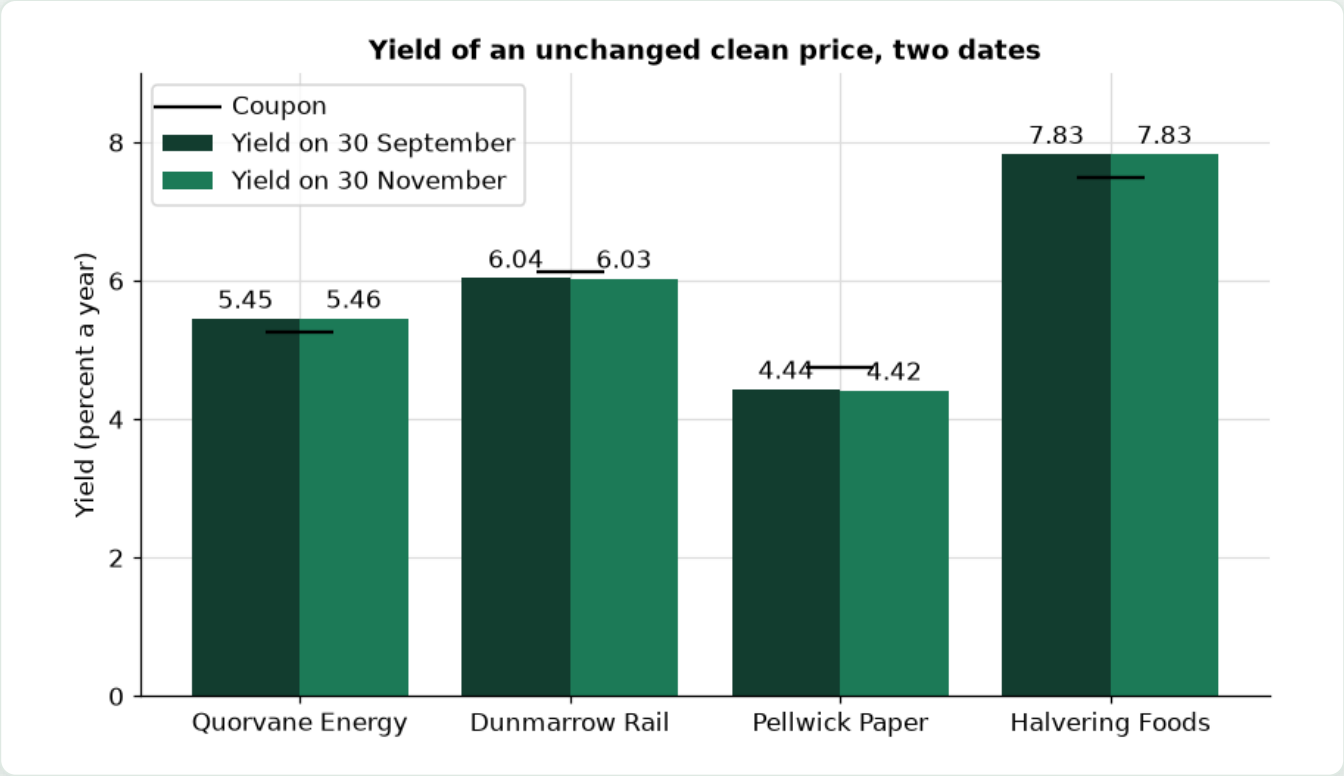
6.5 percent coupon, settled 15 December 2026, maturing 28 February 2027.

Step	Result
PRICE at 5 percent	100.289734
YIELD of that price	5.1370 percent
price at that yield	100.260880

PRICE uses E - A = 75 days. YIELD counts DSR on 30/360: 73 days.

Each does what it documents. bondmath follows Excel in both.

Yields of an unchanged price, drawn



Below par, the yield sits above the coupon. Above par, below it. Two months on, each moves by hundredths of a percent.

A description of the formula on an unchanged price, not a forecast.

Invoice amounts for the September blotter

374

trades, clean price and principal only

1,809,789.22

dollars of accrued at each settle_date

119,283,538.57

dollars invoiced: principal + accrued

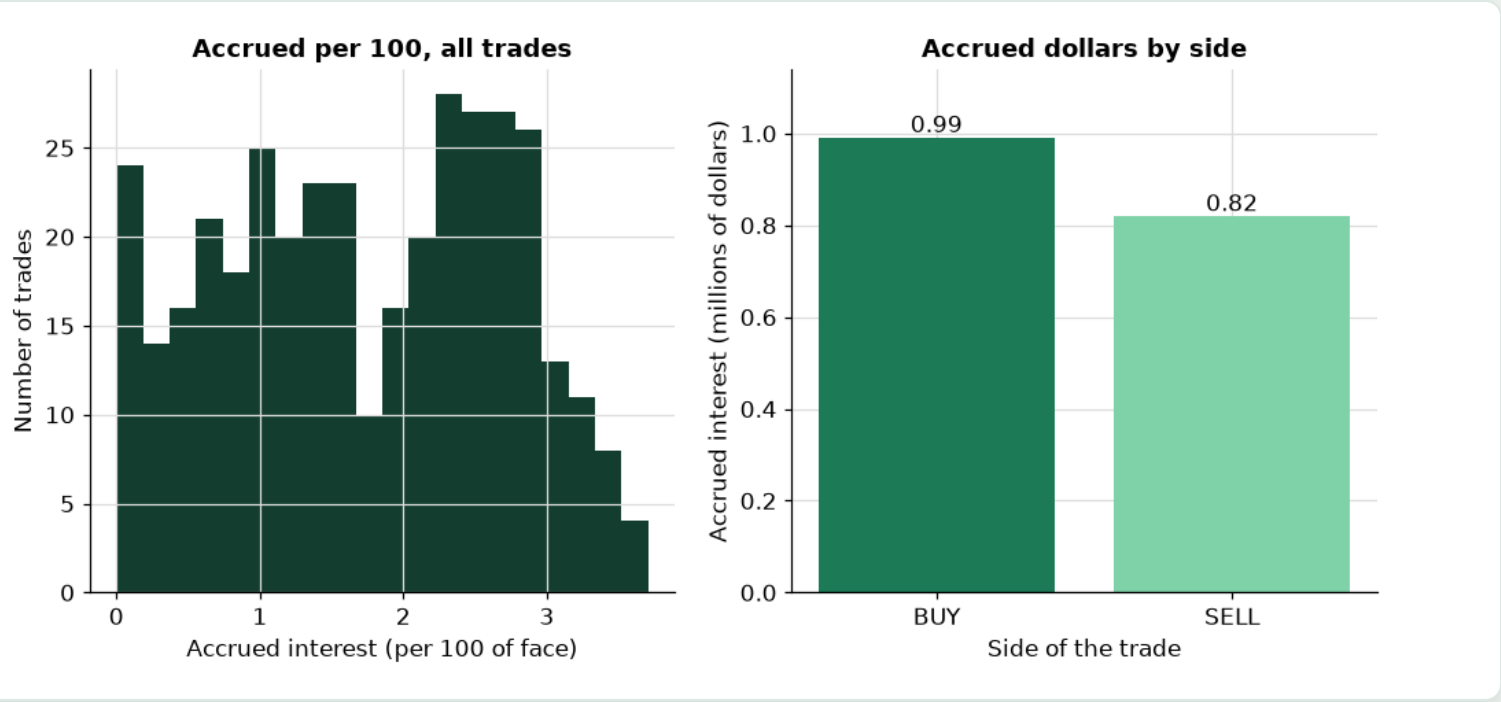
Side	Trades	Principal	Accrued	Invoice
BUY (desk pays)	197	56,556,005.45	990,620.95	57,546,626.40
SELL (desk receives)	177	60,917,743.90	819,168.26	61,736,912.16

Tested in Excel for every trade: face x (price + accrued formula) / 100.

2 trades settled on a coupon date: nothing accrued. Most accrued: T0151, 57,300.35 dollars on 2,500,000 of face.

Accrued is not a cost or a gain. A cash reconciliation on principal alone misses it.

The blotter's accrued interest, drawn



Accrued per 100 runs from zero up to just under a full coupon. The dollars follow the face traded.

Today's Excel, every formula tested

Job	Excel (tested)	Returns
Clean price	`=PRICE(DATE(2026,11,30),DATE(2031,9,30),5.25/100,5.4/100,100,2,0)`	99.361551
Accrued	`=100*5.25/100/2*COUPDAYBS(...)/COUPDAYS(...)`	0.875000
Dirty price	`=PRICE(...)+100*5.25/100/2*COUPDAYBS(...)/COUPDAYS(...)`	100.236551
Written out	`=IF(...)/(1+5.4/100/2)^($B$4+A6)` down, then `=SUM(B6:B15)`	100.236551
One cell	`=-PV(5.4/100/2,COUPNUM(...),5.25/2,100)*(1+5.4/100/2)^(A/E)`	100.236551
Final period	`=PRICE(DATE(2026,9,30),DATE(2027,2,15),5/100,6/100,100,2,0)`	99.619499
Yield	`=YIELD(DATE(2026,11,30),DATE(2031,9,30),5.25/100,99.125,100,2,0)*100`	5.456532

PRICE returns clean and YIELD takes clean. The dirty price you build yourself: add the accrued.

git revert: undo a commit, keep the history

The colleague's commit

```
-   dirty = (100 + coupon) / (1 + w * period_yield)
+   dirty = (100 + coupon) / (1 + period_yield) ** w
```

pytest: 3 failed, each on edge_final_period. The last commit is the broken file, so restore cannot help.

git revert --no-edit HEAD, then git log --oneline

```
abc1234 Revert "Discount the final period like every other period"
abc1234 Discount the final period like every other period
abc1234 Add price, dirty_price, and yield_to_maturity, tested aga...
abc1234 Start of day 8: bondmath.py and test_bondmath.py as day 7...
```

restore undoes a change you have not committed. revert undoes one you have, and keeps the record.

The revert is a new commit that does the opposite of the old one. Never rewrite a history someone may have copied; add to it.

Bring it home: paste, test, commit, revert

1

Your folder

Set-Location to my_bondmath, in your projects folder.

2

Paste

Two module cells and four test cells at the end of your files, without the %%writefile line.

3

Test

python -m pytest -q: 23 passed, or 30 with the exercises of days 1 to 7.

4

Commit

git diff --stat, git add, git commit with a subject and a body.

5

Revert

Make the colleague's change, commit -am, see 3 fail, git revert --no-edit HEAD.

```
git commit -am "..."   python -m pytest -q --tb=no   git revert --no-edit HEAD
```

In Colab: download bondmath.py and test_bondmath.py from the work folder.

The AI check

The docstring is the prompt. The code that comes back runs, and has one convention error.

1

Docstring

Units, convention, and the Excel formula: what you asked for.

→

2

Code back

Written for the course in the style of an AI answer. Read it.

→

3

Test first

Assert against Excel's reference rows. Run it: it fails.

→

4

Fix

Find the convention it broke, fix it, run the test again.

Today: price returns the dirty price as the clean price: the accrued is never taken off.

Code that runs is not the same as code that is right.

What you can do now

- Tell the quote (clean) from the invoice (dirty = clean + accrued)
- Discount over part of a period, w = DSC / E, in Python and in Excel
- price, dirty_price, yield_to_maturity, matching PRICE and YIELD
- Price the final coupon period with simple interest, as Excel does
- Invoice a trade blotter; undo a commit with git revert

Week 2, Day 4 Duration and DV01: how much the price moves when the yield moves